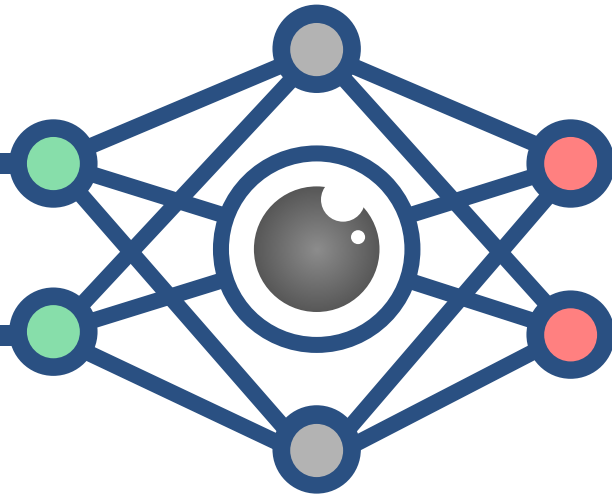


CS3485

Deep Learning for Computer Vision



Lec 1: Introduction to Computer Vision

Course Staff

■ **Instructor:** Jeova Farias

(can call me Jeova or Prof. Farias)

- Email: j.farias@bowdoin.edu
- Office Hours: Tues & Thurs 4:30-5:30pm at VAC 308 (or by appointment or **just come by!**)

■ **Instructor Bio:**

- PhD, Computer Engineering. (Brown University),
- MSc, Applied Math (Brown University), CS (University of Nice, France).

■ **Research interests:**

- Statistical Machine Learning and Discrete Optimization,
- Image Processing and Computer Vision.
- More recently: Unsupervised Deep Learning.

Contact me if you are interested in research in the field!



Learning Objectives

- At of the course, the students are expected to:
 - a. Understand what **Computer Vision** is and how it's been applied in the industry and academia.
 - b. Know what neural networks are and how that led to the development of **Deep Learning**.
 - c. Have a concise knowledge of how Deep Learning has developed over the years and the problems it brought solutions to.
 - d. Have enough working skills on **PyTorch** to developer Deep Learning applications for, and more broadly than, Computer Vision.
 - e. Be able to read and understand the recent literature in Deep Learning and Computer Vision and present their content to a wider audience.



What you can expect from this course

- You can expect **organized lectures and assignments**.
- I expect you to participate in our **in-class activities** and be kind to your peers' questions.
- Prerequisites:
 - In this course, we'll use **math** in some classes, especially calculus and linear algebra
 - However, more importantly: **feel free to ask questions about these topics!**
- We'll go through the very basics of Deep Learning, so, for those who may already know some stuff: **be patient** (and also contribute to the class, if possible).
- Some classes will **have faster pace** than others.



Hardware for Deep Learning

- One big disadvantage when doing/teaching Deep Learning: it requires expensive hardware, i.e., Graphics Processing Units (GPUs).
- **Google Colab** is an option for codes in class, homeworks and your final project:
 - Uses Python notebooks (.ipynb).
 - It is free for anyone with a GDrive account,
 - Provides GPUs for us to run our codes on,
- *One problem:* in the free accounts, it may stop providing GPU access after a lot of usage on it.
- We can also use [Bowdoin's HPC](#), but we'll leave it only for projects, for now.
- If you are interested in using it anyway, I'll write the instructions on how to.

Google Colaboratory

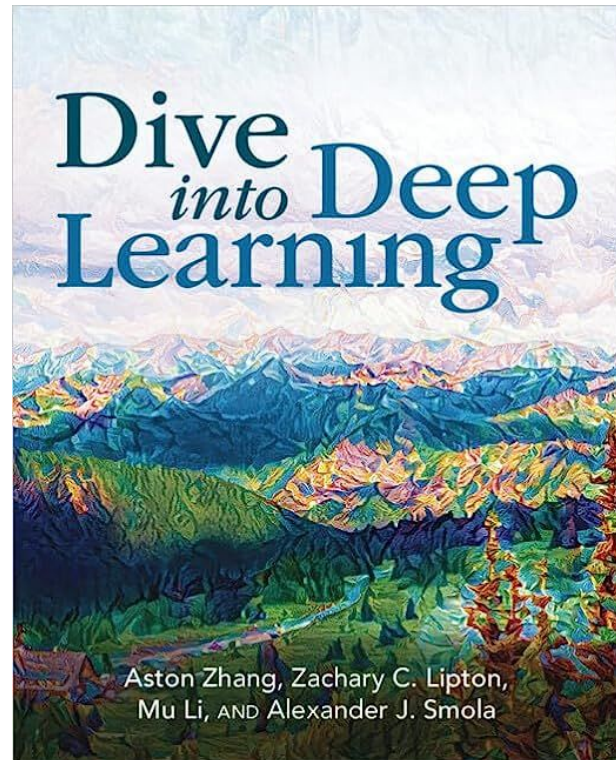


Textbook

- It is hard to find a textbook for this course: the field changes **extremely fast**.
- However, the book we will most use in this course is:
Dive into Deep Learning, by Zhang, Aston, Zachary C. Lipton, Mu Li, and Alexander J. Smola (latest update in 2023).

It's a mostly [online, interactive book](#), whose pdf version is also freely available [online](#).

- It is a great book that covers most of what we'll see here (with code examples) and goes deeply in other topics.
- In practice, you are not required to use it.



Our Website

- We have a website for our course! It's jeovafarias.github.io/Bowdoin-CS3485.
- In it, you shall find:
 - The lecture slides + useful code and papers we discuss in each class;
 - The links to the assignment instructions (their submission will still be via Canvas)
 - The schedule of lectures, exams, breaks and review sessions;
 - Info on the final project.
 - The course syllabus and other materials.
- Despite having a website, we **will use canvas** for some activities.



Course Description

Computer Vision has become ubiquitous in our society, image searches to self-driving cars. On the other hand, Deep learning has shaken the world of artificial intelligence in recent years. Most of these developments greatly advanced the performance of state-of-the-art visual recognition systems, which put Computer Vision at the epicenter of most technological progress from the past decade. In this context, this course aims at providing a consistent exploration of how deep learning started to its most recent achievements, always using Computer Vision tasks as their main application, historically or practically. During the course, we'll also understand many of the main computer vision problems and use them as cases for the introduction of various deep learning related problems. Finally, this course hopes to give students working knowledge of PyTorch, one of the main deep learning frameworks, and prepare them for future industrial and academic careers in the field.

Pre-requisites

Basics of Multivariate Calculus and Linear Algebra (see Materials for some references).

Instructor



Jeova Farias

Course Details

When/Where:

Tuesdays and Thursdays from 8:530p to 9:55p at Searles 223 and at this Zoom link (available when asked)

Office Hours:

Tuesdays and Thursdays from 4-5p and Fridays from 9-11a at VAC 308.

Final grade

■ Labs (20 %):

- An **assignment** released every Thursday, 6-7 total, every 1-2 weeks.
- You'll be given **4 late days for the semester** (budget your days wisely)
- Deliverables will be reports **in Latex** (or else, for 80% of the grade), starting from Lab 2.
- (*Tentative*) Students will pair-up in each lab and deliverable. *I may change this strategy later.*

■ Quizzes (40 %):

- We'll have weekly quizzes, taking place on the last 15 min of our Thursday classes, starting next week. Check our [schedule](#) for more details.

■ Final Project (30 %):

- **Theme:** (1) an new application or model that uses deep learning **or** (2) a review of two or more papers with code **or** (3) an complete piece of software that uses AI in some manner
- **Deliverables:** a **proposal**, code and presentation. Teams of 2-4 people.

■ Participation (10 %):

- Given in two grades one after the break and another at the end of the semester.

Note about final weeks + AI

- There won't be any of my lectures after the last quiz (hopefully in April 21).

Instead I hope to:

- Have a guest lecture from the industry and/or academia.
- Have you guys focus on your final projects.

- In those days, attendance is **mandatory** and it will heavily count towards your final participation grade.
- For those who took AI with me in the past:
 - The first few lectures will be very similar to the last ones last semester.
 - This ensures everybody is “on the same page”.

	11/26/2026 Thursday	THANKSGIVING BREAK!	No lecture.
Guest Lecture	12/01/2026 Tuesday	Guest No. 1	Mandatory student presence.
Project	12/03/2026 Thursday	Work on final projects.	No lecture. Students will work on final projects. Mandatory student presence. Instructor will be there to help students out with their projects!
Project	12/08/2026 Tuesday	Work on final projects.	No lecture. Students will work on final projects. Mandatory student presence. Instructor will be there to help students out with their projects!
Project	12/10/2026 Thursday	Work on final projects.	No lecture. Students will work on final projects. Mandatory student presence. Instructor will be there to help students out with their projects!

GenAI in this course

- AI is a great powerful tool, but with great tools come great responsibility:
 - Many studies show how AI misuse in learning is **very detrimental** to students.
 - In CS, a lot of what we learn depend on trying to solve problems by ourselves and you are handed out solutions you **simply do not learn**.
 - Make sure to **avoid AI misuse** so you can benefit the most (and have the most fun) from this course.
- Here's the main guideline on using Gen AI:

Always acknowledge what generative AI tools were used to help you complete assignments

But also check our complete [AI Policy](#).

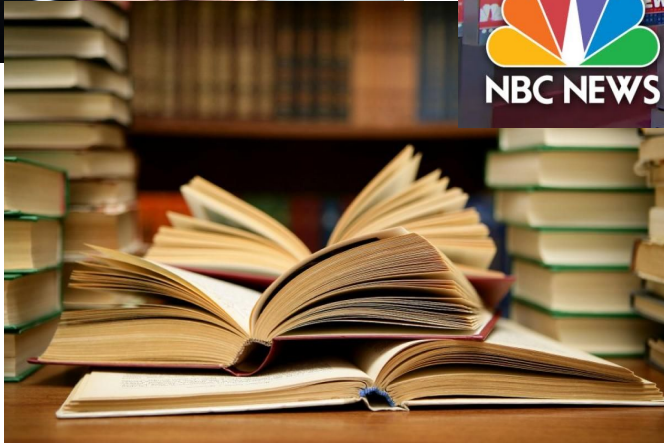


Final Thoughts

- **What else counts as participation?**
 - **Asking** and **answering** questions in class (very important!) and attending office hours.
 - **Not being consistently late to class** or disrupting the class environment
 - I'll assess that twice in the semester:
 - **One right after the Fall break**
 - **Another at the end of the semester**
- The lectures will be primarily **in person**, but I can live stream them if *any student* (even if just one) needs it (just let me know a few hours before it).
- The all of this info (as a PDF) is on our **Syllabus** and [website](#) . **Make sure to read it!**



Let's get started: Data and the modern world



A DAY IN DATA

The exponential growth of data is undisputed, but the numbers behind this explosion – fuelled by internet of things and the use of connected devices – are hard to comprehend, particularly when looked at in the context of one day

500m

tweets are sent
every day
Twitter



4PB

of data created by
Facebook, including

350m photos
100m hours of video
watch time
Facebook Research

294bn

billion emails are sent
Radical Group

320bn

emails to be sent
each day by 2021

306bn

emails to be sent
each day by 2020

3.9bn

people use emails

4TB

of data produced by a connected car
Intel

ACCUMULATED DIGITAL UNIVERSE OF DATA

4.4ZB

PwC

2013

44ZB

2020

DEMYSTIFYING DATA UNITS

From the more familiar 'bit' or 'megabyte', larger units of measurement are more frequently being used to explain the masses of data

Unit	Value	Size
b bit	0 or 1	1/8 of a byte
B byte	8 bits	1 byte
KB kilobyte	1,000 bytes	1,000 bytes
MB megabyte	1,000 ³ bytes	1,000,000 bytes
GB gigabyte	1,000 ³ bytes	1,000,000,000 bytes
TB terabyte	1,000 ³ bytes	1,000,000,000,000 bytes
PB petabyte	1,000 ³ bytes	1,000,000,000,000,000 bytes
EB exabyte	1,000 ³ bytes	1,000,000,000,000,000,000 bytes
ZB zettabyte	1,000 ³ bytes	1,000,000,000,000,000,000,000 bytes
YB yottabyte	1,000 ³ bytes	1,000,000,000,000,000,000,000,000 bytes

*A lowercase 'b' is used as an abbreviation for bits, while an uppercase 'B' represents bytes.

65bn

messages sent over WhatsApp and
two billion minutes of voice and
video calls made
Facebook

Searches made a day **5bn**
Searches made
a day from Google **3.5bn**

Smart Insights

463EB

of data will be created every day by 2025
IDC

95m

photos and videos are
shared on Instagram
Instagram Business

28PB

to be generated from wearable
devices by 2020
Statista

The richest data

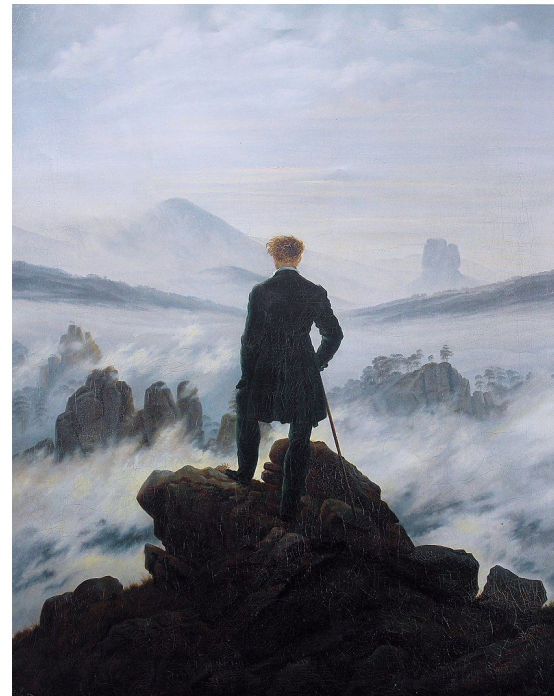
- Arguably, the visual data (image and videos) is the **richest** among all types of data!
- In a simple image, we have have so much information:



1. A couple is drinking wine during the day.
2. The man has a beard, long hair and is wearing a white shirt.
3. The woman also has long hair and is wearing a typical French cap.
4. They seem in love with each other.

What else?

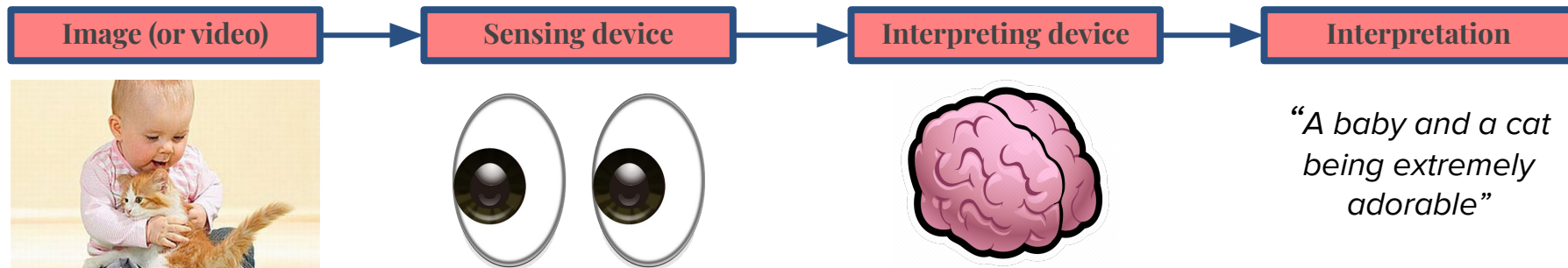
- The visual data is so rich that we perceive **up to 80%** of all impressions by means of our sight.
- **The vision system is our main door to the world!**



Caspar David Friedrich's Wanderer
above the Sea of Fog (1817)

(Oversimplified) Vision pipeline

- In simple terms, our visual system proceeds as following when it sees an image:



- And it works very well: on average, we only need 150ms to detect whether there is an animal in a picture!

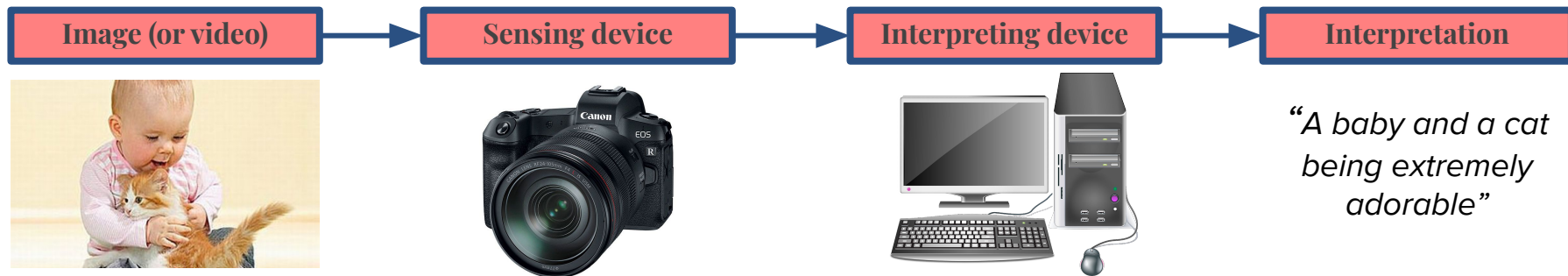


Computer Vision

- As Computer Scientists, we aim to do **Computer Vision**:

Computer Vision (CV) is the study of how computers can gain high-level understanding from **visual data**, such as images and videos.

- Here, we seek to **understand and automate** tasks proper to the human visual system.
- Our *desired* pipeline is:



- Computer Vision is then about **finding the right algorithms** for the interpreting device.

Easy peasy

- At first, Computer Vision seems like an easy problem to solve: “*we humans do it so easily since we were children*”!
- It seemed so easy that its study started with an MIT **undergraduate summer project** in 1966.
- The goal was very “modest”: find a model that rightly mimics our visual system.
- Unfortunately (or fortunately!) vision is a much harder problem to solve, and one summer wasn’t enough time.

MASSACHUSETTS INSTITUTE OF TECHNOLOGY
PROJECT MAC

Artificial Intelligence Group
Vision Memo. No. 100.

July 7, 1966

THE SUMMER VISION PROJECT

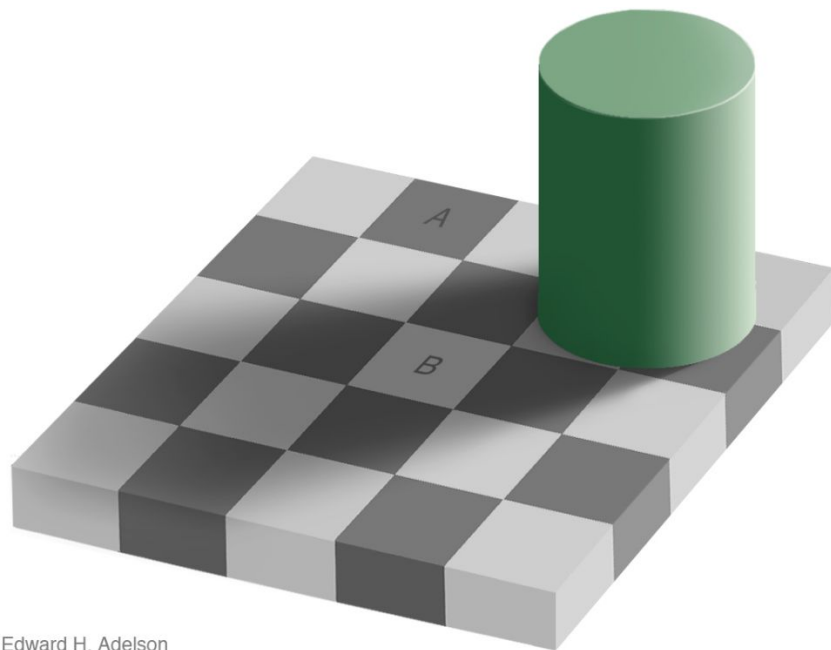
Seymour Papert

The summer vision project is an attempt to use our summer workers effectively in the construction of a significant part of a visual system.

The particular task was chosen partly because it can be segmented into sub-problems which will allow individuals to work independently and yet participate in the construction of a system complex enough to be a real landmark in the development of "pattern recognition".

Example of why vision is complicated

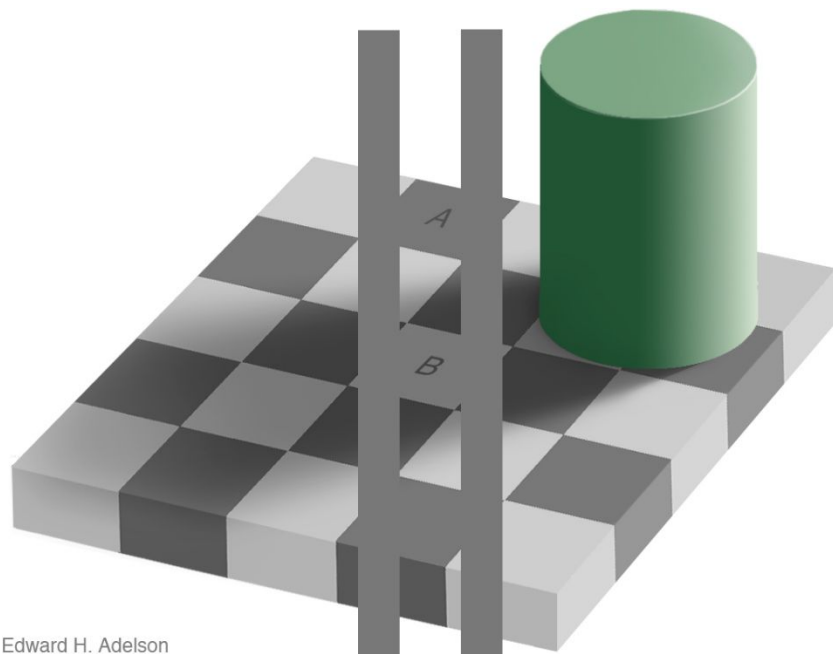
- Which region is darker in the image below, A or B ?



Edward H. Adelson

Example of why vision is complicated

- Believe it or not, but A and B are of the same shade of gray!



Edward H. Adelson

Exercise (*In pairs*)

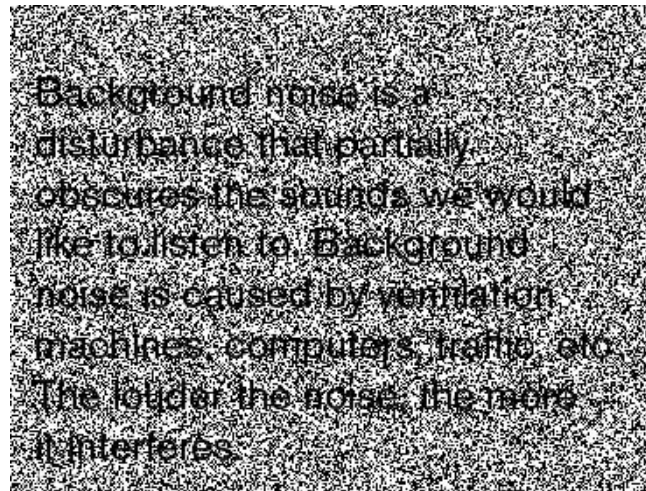
- Answer the questions for the following images:



How many objects are there in this image?



Which object is the largest? What color is the sky?



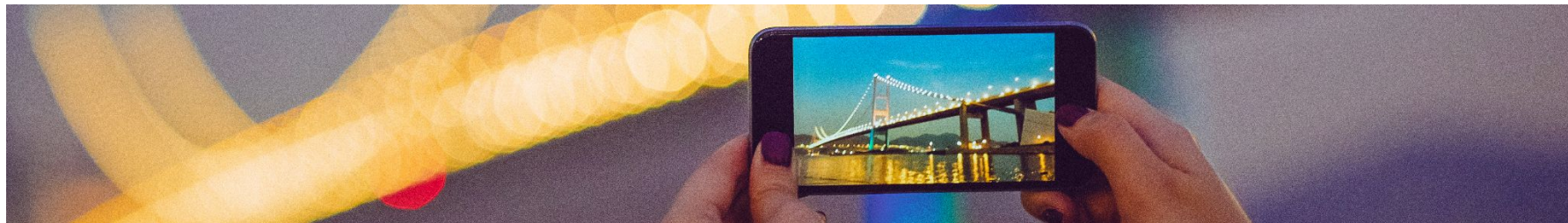
What is written in the above image?

Why vision is complicated

- Computer Vision is hard for many reasons, like:
 - **Our visual system is limited in some tasks:**
 - Our vision is prone to illusions that lead to misinterpretation of certain phenomena.
 - That raises the question of whether we want to necessarily mimic it using CV.
 - **Many vision problems are ambiguous:**
 - There is usually not an objective solution for some problems, as they may depend on human subjective reasoning.
 - That makes the modeling of those problems harder.
 - **Images remove important aspects of the observed scene:**
 - Usually the distance from the camera to the objects is unavailable, making the distance between the objects in a scene look wrong.
 - Also, some color information may not match to what the real scene's colors are.
 - **The sensing device is imperfect, and the images are noisy:**
 - It is usually impossible to capture a real world scene without adding noise to it.

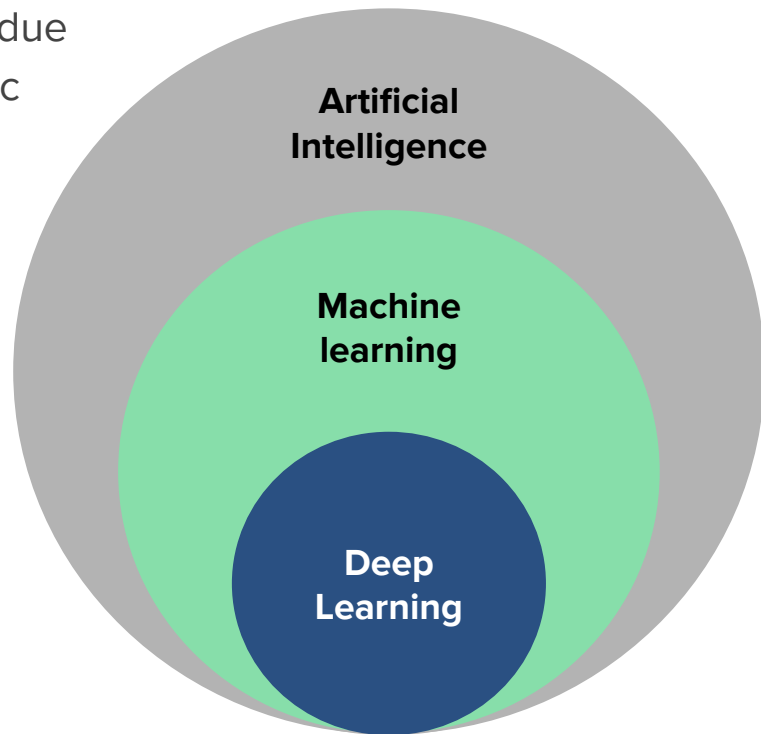
Why is Computer Vision so Important Now

- Despite these issues, technology is progressed at an exponential pace and computer vision solutions found numerous applications in modern society.
- This is due to the following factors:
 - **Computing Power:** Hardware designed for computer vision has become cheaper, faster, better and easily accessible.
 - **Big Data:** Availability of large training data sets due to mobile technology with built-in cameras saturating the world with photos and videos
 - **Open Source:** Availability of new algorithmic frameworks that can take advantage of hardware and software capabilities



Deep Learning and Computer Vision

- Largely, the great success of Computer Vision is due to recent use of **Deep Learning** as an “algorithmic framework” (the interpreting device).
- Based on Neural Networks, it a **supervised machine learning model** “inspired” by the way the brain propagates information.
- Overall it sits right at the core of all the **Artificial Intelligence Revolution** we are currently experiencing
- It was initially used for **image classification**, it has been extended to solve problems way beyond Computer Vision!



Deep Learning in the news

Deep learning helps predict traffic crashes before they happen

A deep model was trained on historical crash data, road maps, satellite imagery, and GPS to enable high-resolution crash maps that could lead to safer roads.

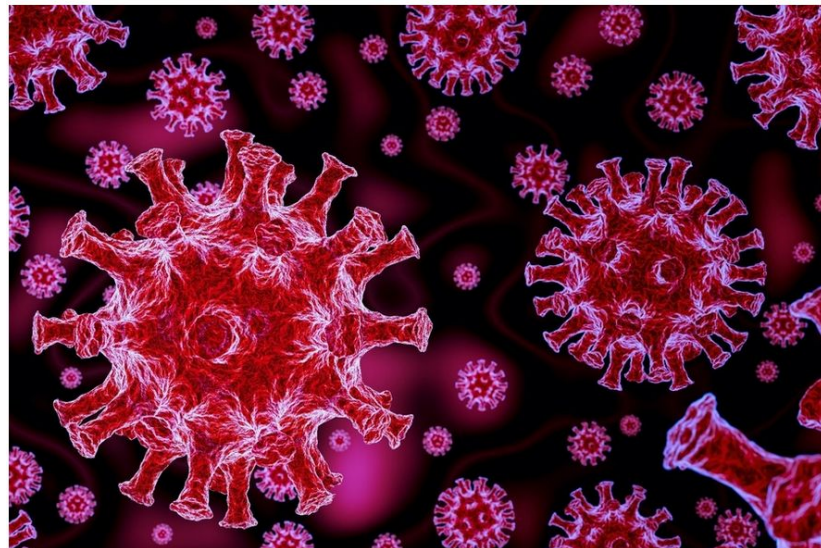
Rachel Gordon | MIT CSAIL
October 12, 2021



Deep learning helps predict new drug combinations to fight Covid-19

Neural network identifies synergistic drug blends for treating viruses like SARS-CoV-2.

Rachel Gordon | MIT CSAIL
September 24, 2021



Deep Learning in the news

NEWS | 09 October 2024

Chemistry Nobel goes to developers of AlphaFold AI that predicts protein structures

Study finds ChatGPT boosts worker productivity for some writing tasks

A new report by MIT researchers highlights the potential of generative AI to help workers with certain writing assignments.

Zach Winn | MIT News Office
July 14, 2023

Editorial | Published: 22 April 2026

Towards a coordinated approach to LLMs in astronomy

[Nature Astronomy](#) 10, 467 (2026) | [Cite this article](#)



[Save article](#)

2569 Accesses | 1 Citations | [Metrics](#)



How to do Deep Learning

- Over the years, many libraries in Python were implemented for developing applications using deep learning.
- Currently, the most famous ones (both open-sourced) are:

Tensorflow



- Created in 2015 by Google Brain department (Version 2.0 launched in 2019).
- Steep learning curve.
- Usually used in pair with Keras, an easy interface for it
- Large community, many tutorials.
- Use to be DL's main library.

Pytorch



- Developed in 2016 by Facebook's AI Research lab.
- Based on a library called Torch, written in Lua,
- Highly "Pythonic", easy to learn.
- Gives you more control.
- Small (but growing) community.
- Research preferable library.

- In our course, we'll use **PyTorch**, and we'll have a lecture solely on it.

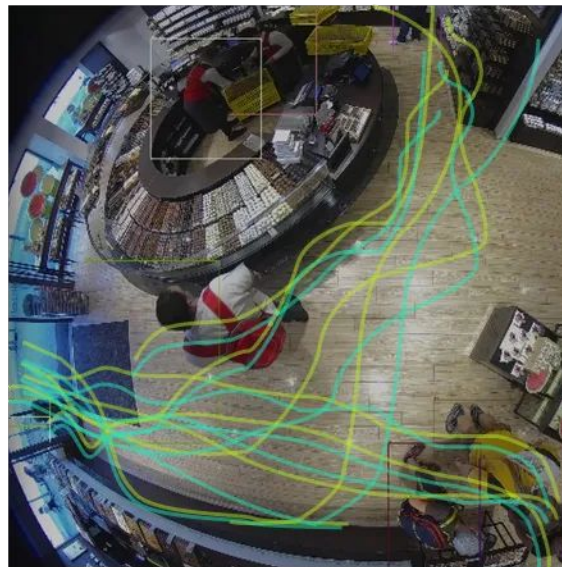
Applications of Computer Vision

- **Shopping** will never be the same without Computer Vision!

Cashierless Shopping



Customer tracking and Flow



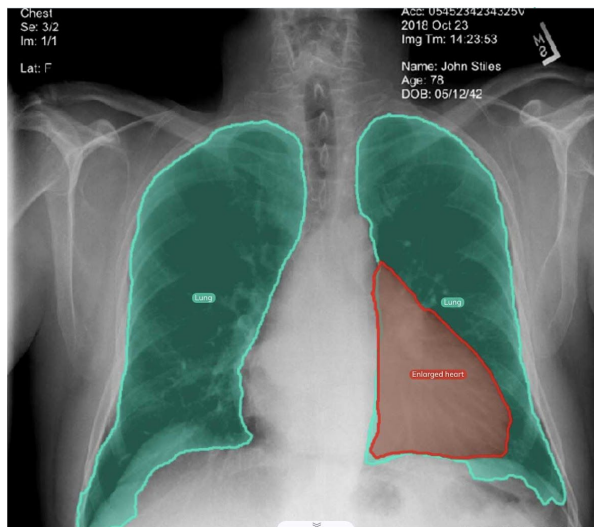
Gaze detection



Applications of Computer Vision

- Without the Computer Vision, **healthcare** professionals would forced to spend hours manually analyzing patient data.

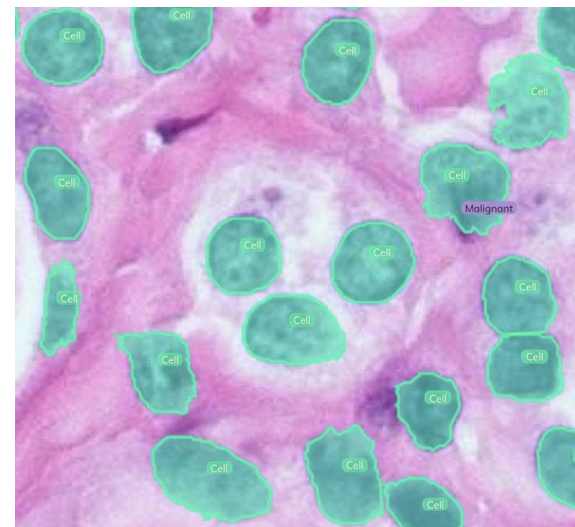
X-Ray image segmentation



MRI understanding



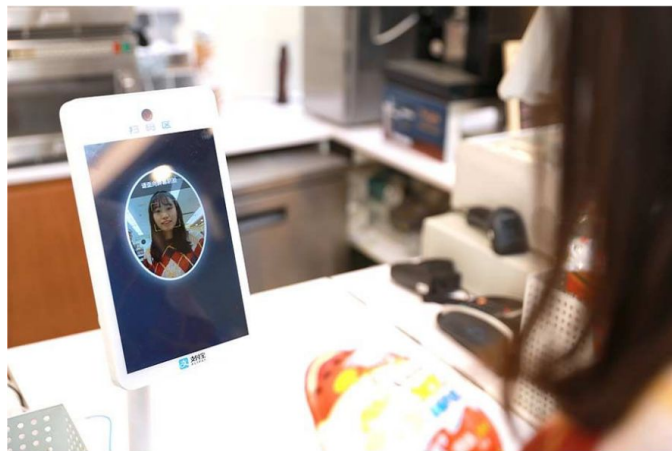
Pathology detection



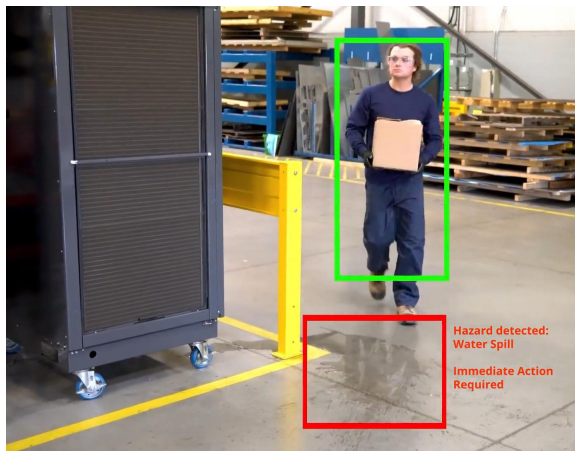
Applications of Computer Vision

- Computer vision enables many possibilities in **security and surveillance**!

Face Recognition



Hazard Detection



Crowd Counting/Control



- Good news for public safety: helping police and first responders more easily spot crimes and accidents.

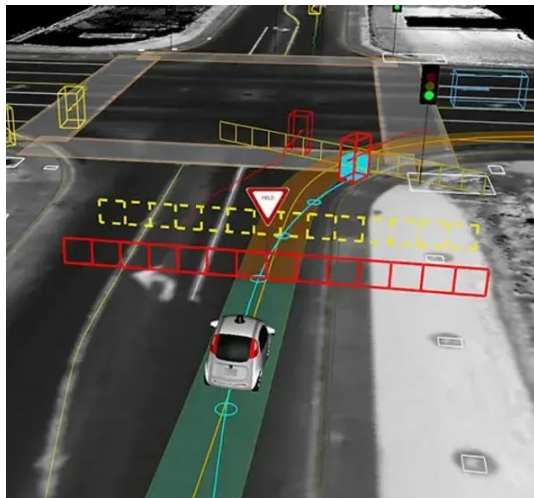
Applications of Computer Vision

- The future of **self-driving cars** depends on Computer Vision:

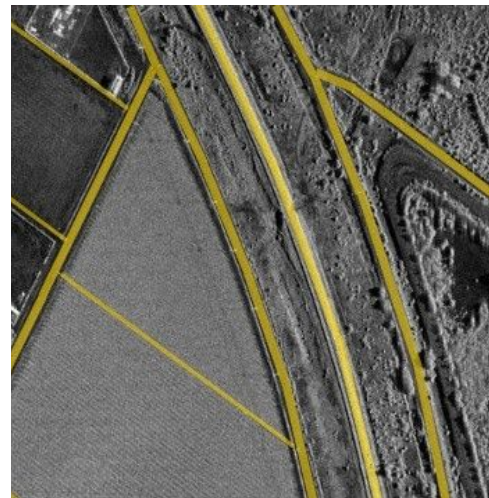
Car and pedestrian detection



Trajectory Planning



Road Detection

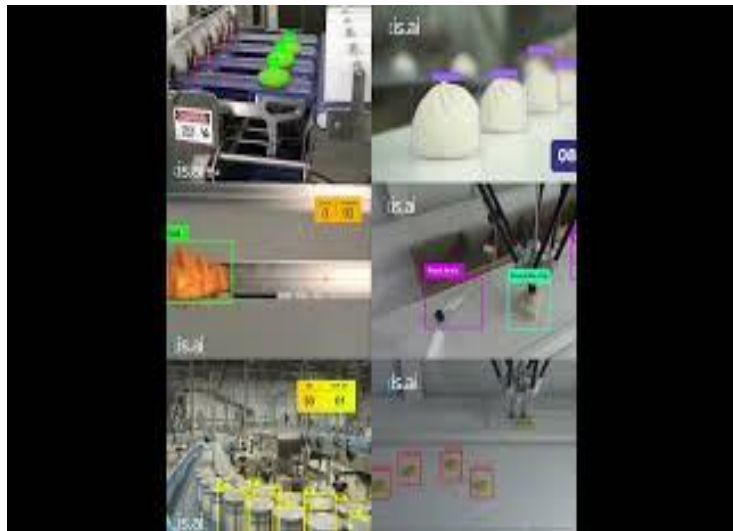


- And this all is already implemented on Tesla cars.

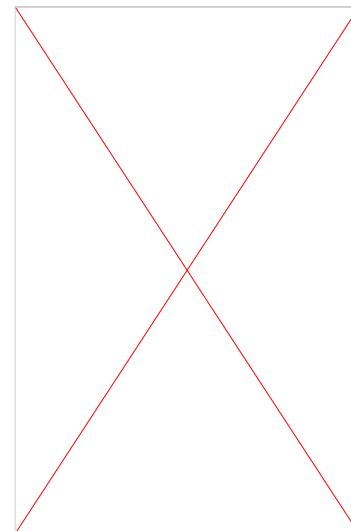
Applications of Computer Vision

- ... and of robotics:

Manufacturing



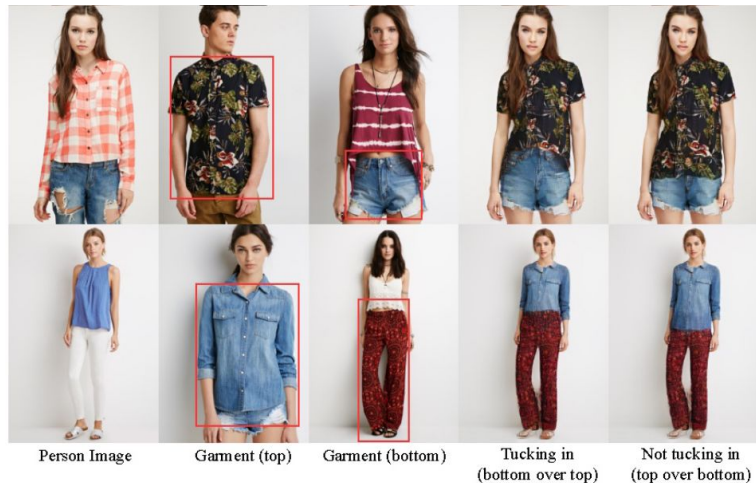
Movement Planning



Applications of Computer Vision

- Also, many applications in the **fashion** industry:

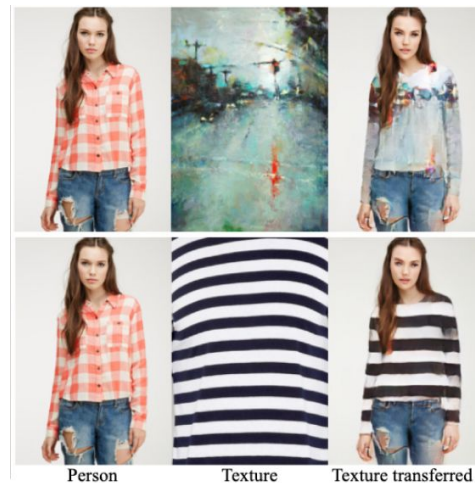
Virtual Try-on



Garment Layering



Clothing editing



Source and more: cuiaiyu.github.io/dressing-in-order/

Applications of Computer Vision

- And it has overflown the **digital art** industry recently!

Style Transfer



Image Generation by Prompt



Image generated with the prompt: "an impressionist oil painting of a Canadian man riding a moose through a forest of maple trees"

Applications of Computer Vision

- And now Vision also intersects Language!

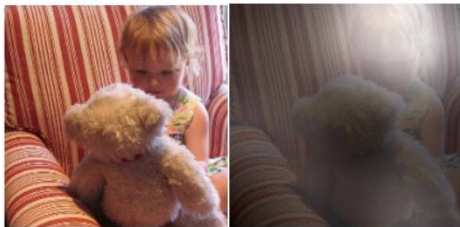
Image Captioning



A woman is throwing a frisbee in a park.



A dog is standing on a hardwood floor.



A little girl sitting on a bed with a teddy bear.



A group of people sitting on a boat in the water.

Image Generation by prompt



"a cute doll writing a letter"



"a bored smoking lizard surrounded by soldiers"



"a parking meter near a graffiti wall"



"a corgi in a field"



"a monkey eating a banana"



"a neapolitan pizza with mozzarella and tomatoes"

Applications of Computer Vision

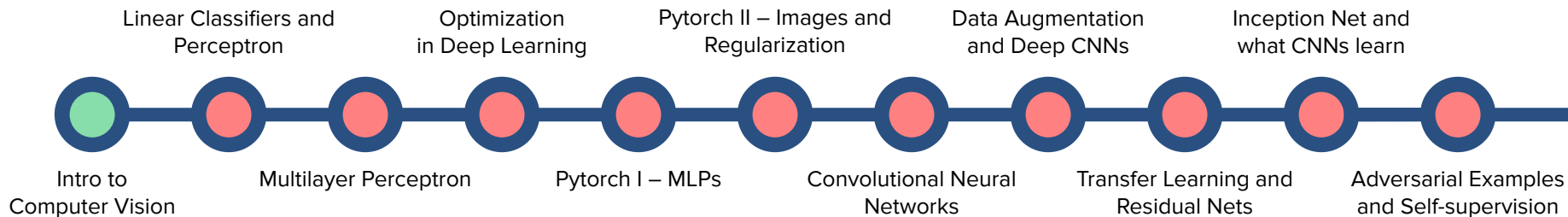
- And obviously, Video Generation!

Video Generation

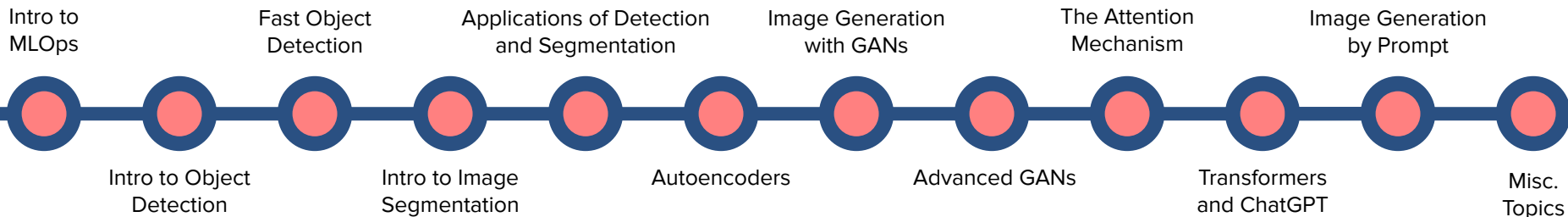


(Tentative) Lecture Roadmap

Basics of Deep Learning



Deep Learning and Computer Vision in Practice



Video: *The Deep Learning Revolution*

